

1. What is C?

C is a general-purpose, procedural programming language used for system and application development. It is fast, efficient, and widely used for operating systems and embedded software.

2. What are the features of C?

C is simple, portable, structured, and efficient. It supports low-level memory access and modular programming.

3. What is a variable?

A variable is a named memory location used to store data. Its value can change during program execution.

4. What are data types in C?

Data types define what kind of value a variable can store. Common types are int, char, float, and double.

5. What is a constant?

A constant is a fixed value that cannot be modified during program execution. Constants improve code readability.

6. What is an operator?

Operators perform operations on variables and values. Examples include arithmetic, relational, and logical operators.

7. Difference between = and ==?

= assigns a value to a variable. == compares two values and returns true if they are equal.

8. What is a function?

A function is a reusable block of code that performs a specific task. It improves modularity and code reuse.

9. What is recursion?

Recursion is when a function calls itself to solve a smaller version of a problem. It ends when a base condition is met.

10. What is an array?

An array stores multiple elements of the same data type in contiguous memory. Elements are accessed by index.

11. What is a pointer?

A pointer stores the address of another variable. It enables efficient memory access and dynamic allocation.

12. What is a NULL pointer?

A NULL pointer points to no valid memory location. It is used to indicate an uninitialized or empty pointer.

13. What is a string?

A string is a character array ending with the null character '\0'. It is used to store text.

14. What is a structure?

A structure groups variables of different data types under one name. It represents related data together.

15. What is a union?

A union allows different members to share the same memory location. Only one member holds a valid value at a time.

16. What is enum?

An enum is a user-defined type with named integer constants. It makes code easier to read.

17. Storage classes?

Storage classes define a variable's scope, lifetime, and visibility. Examples are auto, static, extern, and register.

18. static keyword?

A static variable retains its value between function calls. A static function is visible only within its source file.

19. extern keyword?

extern declares a variable or function defined in another file. It enables sharing global variables.

20. Dynamic memory allocation?

Memory is allocated at runtime using malloc, calloc, or realloc and released with free. It provides flexible memory usage.

21. Memory leak?

A memory leak occurs when allocated memory is not freed. It wastes memory and can reduce performance.

22. Header file?

A header file contains function declarations, macros, and definitions. It is included using #include.

23. Preprocessor?

The preprocessor handles directives before compilation. Examples include #include and #define.

24. Macro?

A macro is a named code fragment created with #define. It is replaced before compilation.

25. while vs do-while?

while checks the condition before execution. do-while executes the loop once before checking.

26. break statement?

break immediately exits a loop or switch statement. Control moves to the next statement.

27. continue statement?

continue skips the current iteration and proceeds to the next iteration of the loop.

28. File in C?

A file stores data permanently. C provides functions like fopen, fclose, fread, and fwrite for file handling.

29. Command-line arguments?

Arguments are passed to main() through argc and argv. They allow input when running a program.

30. Why is C middle-level?

C supports both low-level memory manipulation and high-level programming constructs. This makes it versatile for many applications.